Sports and Biometrics: Indian Summer Olympic Athletes

CS 215: Data Analysis and Interpretation — Final Project Report (Checkpoint 2)

4 November 2026

1. Meta Information

1. **Open project title:** Sports and Biometrics: Indian Summer Olympic Athletes.
2. **Group ID (string):** "13".
3. **Mentor:** Mayank Jain.
4. **Members and roll numbers:** Geetha (25B0915); Manohar (25B1043); Nemmadi Aakash (25B1061); Teenu (25B1091).
5. **Honor Code and Discussions:** We, Group 13, pledge on our honor that we have fully disclosed and cited all external resources, tools, and discussions utilized throughout this project. We have not engaged in uncredited consultations outside our group.
6. **LLM Usage Links:**
   - https://chat.z.ai/s/08f8483f-d802-4ece-9f5d-cdcf41036190
   - https://share.gemini.google/Cxr4peZYhmIZ
   - https://chatgpt.com/s/t_6aeb6af11b3c81918639
7. **PPT Pdf:** https://drive.google.com/file/d/1jg599_6S3e6oSNV3zdhCQIso32srWjWk/view?usp=drivesdk
8. **Video presentation:** https://drive.google.com/file/d/1hnXB5_4VtuO93WlrKYCMuT85IFcvNlVb/view?usp=drivesdk
9. **GitHub repository:** https://github.com/getsen18/Sports_biometric_DAI.
10. **GitHub handles and initial commit dates:**

GitHub handles and initial commit dates:

Member (roll)	GitHub handle	First commit (YYYY-MM-DD)
Geetha (25B0915)	getsen18	2026-11-04
Manohar (25B1043)	Manohar-07-git	2026-10-21
Nemmadi Aakash (25B1061)	Aakash-nemmadi32	2026-11-04
Teenu (25B1091)	teenuanand456	2026-11-04

11. **Contribution Breakdown:** Unequal
    - 25B0915: 100%
    - 25B1043: 100%
    - 25B1061: 100%
    - 25B1091: 100%
    - LLM: 25%

2. Technical Content

2.1 Research problem and analytical questions

Indian Olympic athletes do not form one homogeneous physical population. Different sporting demands, sex composition and historical participation patterns may all affect observed age and body measurements. We addressed three related questions: **(Q1)** How do age, height, weight and derived BMI differ across sports-based functional archetypes, including sex-wise comparisons? **(Q2)** Does the association between height and weight differ between archetypes after controlling for sex? **(Q3)** Is Olympic year associated with athlete weight when height, sex and sporting composition are taken into account? The aim is descriptive and associational inference, not an estimate of the causal effect of playing a sport or of time.

2.2 Data provenance, coverage and selection

The primary source is the publicly accessible Kaggle *120 Years of Olympic History* athlete-events dataset (`https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results`); Olympedia was a possible cross-check for athlete identities (`https://www.olympedia.org/`). Filtering for Summer Games and NOC = IND, and removing exact duplicate records, gave 1,378 entries. Requiring observed height and weight reduced the data to 835 participation/event records spanning 1936–2016; these are not 835 distinct individuals. The Checkpoint 1 extract contains 12 columns: `ID`, `Name`, `Sex`, `Age`, `Height`, `Weight`, `Year`, `City`, `Sport`, `Event`, `Medal`, `Clean_Name`. It spans 21 sports, 137 events and 18 host cities. Sex is coded in two recorded categories. The project summary reports 513 distinct cleaned-name values, while the later athlete-ID-based workflow counts 522 athletes; these are different identifiers and should not be treated as equivalent.

2.3 Cleaning, feature engineering and analysis units

The pipeline retained complete height-weight pairs, sorted by athlete ID and edition, and filled missing ages with sport-by-sex median values (overall-median fallback). Missing medals were labelled "No Medal"; event names and team labels were standardized, a cleaned name field was created, and remaining missingness was checked. BMI was constructed as $\text{BMI} = W/(H/100)^2$, where W is kilograms and H centimetres. Sports and event types were grouped into seven investigator-defined archetypes: Combat, Endurance, Precision, Racket Sports, Speed & Power, Team Sports and Technical/Mixed. These group labels are interpretations rather than original dataset attributes.

The unit of analysis was selected to match each question. Q1 used 737 athlete-year-archetype observations after deduplication within those keys; Q2 started with 732 athlete-year records and excluded five multi-archetype assignments for its 727-record common modelling sample. Q3 used the full 835 event-level entries, thereby weighting repeated event participation. This distinction affects both estimates and uncertainty.

2.4 Software and statistical approach

The project used Python and Jupyter notebooks, with pandas/NumPy for wrangling, Matplotlib/Seaborn for plots, and statsmodels/SciPy-style statistical functionality for regression, inference and diagnostics. For Q1, we compared means, dispersion, boxplots, histograms, sex-stratified summaries and scatterplots; BMI was treated as height-adjusted mass, not a measurement of body-fat percentage. For Q2, we fitted ordinary least squares (OLS), moving from $W_i = \beta_0 + \beta_H H_i + \epsilon_i$ to an adjusted interaction model:

$$W_i = \beta_0 + \beta_H H_i + \theta F_i + \sum_{j=1}^{6} \gamma_j A_{ij} + \sum_{j=1}^{6} \delta_j (H_{c,i} A_{ij}) + \epsilon_i$$

Here H_c denotes centred height, F a female indicator, and A_j archetype indicators (male/Endurance reference). A partial F-test assessed $H_0 : \delta_1 = \cdots = \delta_6 = 0$. Residual and Q-Q plots probed assumptions; an athlete-grouped 80/20 train/test split avoided putting the same athlete in both partitions. For Q3, regressions of weight on year and height were compared before and after sex/archetype adjustment, using athlete-clustered standard errors.

2. Technical Content (continued)

2.5 India relevance and descriptive results

Every included observation represents an athlete recorded for India at the Summer Olympics; thus the project is directly about India's national sporting representation, rather than extrapolating patterns from international athletes. The cleaned event-level sample has mean age 25.64 years (SD 5.39), mean height 171.70 cm (SD 9.60), and mean weight 67.70 kg (SD 14.37). Archetype-level summaries indicate real descriptive heterogeneity:

Archetype	Q1 obs.	Mean age (y)	Height (cm)	Weight (kg)
Combat	110	24.10	169.64	68.85
Endurance	56	26.21	172.54	61.04
Precision	97	29.21	171.80	71.57
Racket Sports	61	26.20	176.66	70.98
Speed & Power	139	24.90	171.25	68.06
Team Sports	245	24.74	171.80	66.20
Technical/Mixed	29	28.90	172.59	65.76

Precision had the highest mean age (29.21 years), Racket Sports the highest mean height (176.66 cm), and Endurance the lowest mean weight (61.04 kg) and BMI (20.36). Unequal group sizes, including just 29 Technical/Mixed observations, require caution about rank comparisons. Sex-specific plots were used to distinguish archetype patterns from differences in male/female representation.

2.6 Height-weight regression: model comparison and formal test

On the initial 732 athlete-year sample, Pearson's height-weight correlation was $r = 0.698$. The baseline fit was $\hat{W} = -107.782 + 1.021H$ with $R^2 \approx 0.487$. Although taller athletes tended to weigh more, the pooled residual means differed by archetype: Endurance -7.27 kg, Combat $+3.36$ kg and Precision $+3.87$ kg. Thus a single line exhibited systematic subgroup errors.

On the same 727 records, adding sex, archetype and height-archetype interactions improved R^2 from about 0.487 to 0.576. Estimated height slopes ranged from 0.613 kg/cm for Team Sports to 1.261 kg/cm for Speed & Power. The partial interaction test yielded $F(6, 712) = 9.4019$, $p \approx 6.01 \times 10^{-10}$, rejecting a common slope at 5% under standard OLS assumptions. The fitted female coefficient was about -3.83 kg, conditionally on height and archetype. These are conditional associations, not causal or universal physiological differences.

Model validation used 580 training and 147 test records, representing 416 and 105 distinct athletes, respectively. Q-Q plots showed deviations in residual tails. Conventional individual 95% prediction intervals were often wide: a held-out 1992 entry predicted at 53.23 kg had an interval of $[34.28, 72.17]$ kg, illustrating the limits of individual prediction even when average fit improves.

2.7 Olympic-year trends and limitations

Q3 estimated a positive association between Olympic year and recorded weight: 0.71 kg/decade adjusting for height, increasing to 1.00 kg/decade after adding sex and archetype. A men's hockey sensitivity check estimated 1.31 kg/decade, using 198 records from 132 athletes. The year coefficient compares participants across editions; it does not track an individual's weight gain. Athlete-clustered uncertainty was used for Q3, whereas Q2's classical F-test and prediction intervals remain sensitive to repeated observations and unequal/non-normal residual variation. Additional limitations include exclusion of 543 entries without complete height-weight data, historically variable measurement coverage, unequal sport/sex representation, subjective archetype boundaries and event-level weighting. Consequently our claims apply to recorded Indian Summer Olympic participation and should not be generalized uncritically to all Indian athletes.

2.8 Reproducibility and references

The GitHub repository contains cleaning notebooks, Question 1 classification/statistics/plots and report, and the Question 2 modelling notebook/report. Relevant sources are: Kaggle dataset cited above; https://www.olympedia.org/ (potential athlete cross-check); and course project guidelines (CS 215, updated 8 October 2026). Figures in project analysis were generated from the group's data workflow, rather than copied from external articles.

Section 3: Frequently Asked Questions (FAQ)

Q1. Why is Body Mass Index (BMI) potentially misleading when comparing Olympic athletes?

Answer: BMI (kg/m^2) cannot distinguish between lean muscle mass and adiposity. Two athletes with identical BMI values can display drastically divergent physiological profiles (e.g., heavily muscled combat athletes versus lean marathoners). While useful for descriptive population-level standardization, BMI is insufficient to assess athletic fitness.

Q2. Why must we evaluate male and female athletes separately, and could aggregate pooling introduce compositional bias?

Answer: Pooling observations across sexes can confound archetype comparisons if sex ratios differ systematically across sports. An archetype appearing heavier overall may simply have a higher proportion of male competitors rather than sport-specific physical demands. Failing to control for sex introduces confounding bias, where aggregate averages distort the underlying group patterns. Valid inference requires sex-stratified distributions prior to drawing cross-archetype conclusions.

Q3. Why does linear regression minimize squared residuals rather than their algebraic sum?

Answer: For OLS, residuals are $e_i = y_i - \hat{y}_i$. Minimizing their raw sum allows positive and negative errors to cancel symmetrically despite poor fit. Squaring residuals prevents cancellation, penalizes large outliers, and guarantees a unique convex minimum. Our baseline model yields:

$$\widehat{W} = -107.782 + 1.021H$$

where each additional centimeter of height is associated with approximately 1.021 kg greater expected weight.

Q4. What empirical findings justified transitioning from simple linear regression to multiple linear regression?

Answer: Height alone explained only 48.7% ($R^2 = 0.487$) of observed weight variance. Crucially, residual analysis across archetypes revealed systematic bias: the baseline model systematically underestimated combat athletes (mean residual $+3.36$ kg) and overestimated endurance athletes (mean residual -7.27 kg), indicating that sporting archetype and sex capture substantial variance uncaptured by stature alone.

Q5. What is the formal and structural difference between adding archetype indicators versus interaction terms?

Answer: Indicator variables induce parallel intercept shifts while holding the height-to-weight slope constant across sports. Interaction terms (Height $\times$ Archetype) allow the scaling slope itself to vary by discipline. Empirically, slopes diverged widely—yielding 1.261 kg/cm for Speed & Power versus 0.613 kg/cm for Team Sports—demonstrating distinct morphometric scaling behaviors across sports.

Q6. What were the formal hypotheses and empirical findings of our interaction F-test?

Answer: The nested interaction test evaluated:

$$H_0 : \delta_1 = \delta_2 = \cdots = \delta_6 = 0 \quad \text{vs.} \quad H_1 : \exists\, j \text{ s.t. } \delta_j \neq 0$$

The null posited a common height slope across archetypes after adjusting for sex and base intercept differences. The test yielded $F(6, 712) = 9.4019$ with $p \approx 6.01 \times 10^{-10}$, providing decisive evidence to reject a homogeneous slope across athletic archetypes.

Q7. How do confidence intervals and prediction intervals differ in our athletic regression model?

Answer: A confidence interval quantifies uncertainty around the conditional mean $\mathbb{E}[W \mid X]$. A prediction interval also accounts for irreducible individual variance (σ^2). For one held-out athlete, the model predicted $\widehat{W} = 53.23$ kg with a 95% individual prediction interval of $[34.28, 72.17]$ kg, highlighting that aggregate model precision does not eliminate wide individual-level dispersion.

Q8. Why were athlete-clustered standard errors necessary when evaluating multi-Olympic temporal trends?

Answer: Longitudinal Olympic data contains multi-year entries from recurring athletes across Games, violating the *i.i.d.* error assumption. Unadjusted standard errors underestimate parameter variance due to positive intra-athlete serial correlation. Athlete-clustered standard errors correctly account for this repeating variance without artificially overstating statistical precision.

4. Reflection Essay (Optional)

This project changed our understanding of data analysis from a sequence of formulas into a process of asking answerable questions. Basic tools such as groupwise summaries, visualization and regression were familiar in isolation, but applying them to a shared dataset forced us to decide which comparisons were meaningful. For example, the apparent simplicity of a height-weight scatterplot concealed differences across sporting archetypes. Inspecting subgroup residuals led naturally to interaction terms and then to a formal nested-model test. We learnt that a convincing statistical result should emerge from a clear question, transparent modelling decisions and diagnostic checks, not merely from reporting a small p-value.

The messiness of historical Olympic records was equally instructive. We had to choose how to handle missing age and biometric values, standardize event labels, construct interpretable archetypes and decide what counted as an independent observation. Unlike structured homework, none of those choices had a single predetermined answer. Different questions required different analysis units: a cleaned event-level table, an athlete-year representation and an archetype-linked summary. We also learnt to identify what a dataset cannot support. Complete-case filtering may change the population described, BMI is not body composition, and a positive year coefficient does not demonstrate an individual's weight increasing over time.

Working collaboratively strengthened practical skills beyond statistics: using GitHub to exchange notebooks, writing reproducible Python pipelines, maintaining consistent naming conventions, interpreting plots, documenting limitations and translating technical results into a short presentation. The regression work made especially clear why model diagnostics and held-out evaluation matter. Improving R^2 was valuable, but the wide individual prediction intervals reminded us not to oversell a model's precision. We also learnt to treat LLM assistance as an aid for coding or communication that requires independent verification and explicit disclosure.

These habits are relevant to later academic research and data-oriented internships or placements. In real settings, the hardest step is often turning a broad problem into measurable questions while being honest about sampling and uncertainty. If repeating the project, we would define sensitivity checks and observation units earlier, investigate historical missing-data mechanisms more systematically, and reserve additional time for robust uncertainty estimates. The main outcome was not any one coefficient; it was greater confidence in designing an auditable analysis and communicating both what the evidence says and where it stops.